



# Build an Agent Harness with Microsoft Agent Framework and GitHub Copilot SDK

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# WHO AM I?

- Thang Chung
- Solutions Architect at Tiger Tribe (a Heineken company)
- Microsoft Azure MVP
- Focus on Enterprise Architecture, IAM, Scaling AI, Agentic AI
- Coder: <https://github.com/thangchung>
- Worker: <https://www.linkedin.com/in/thangchungatwork/>
- Blogger: <https://dev.to/thangchung>

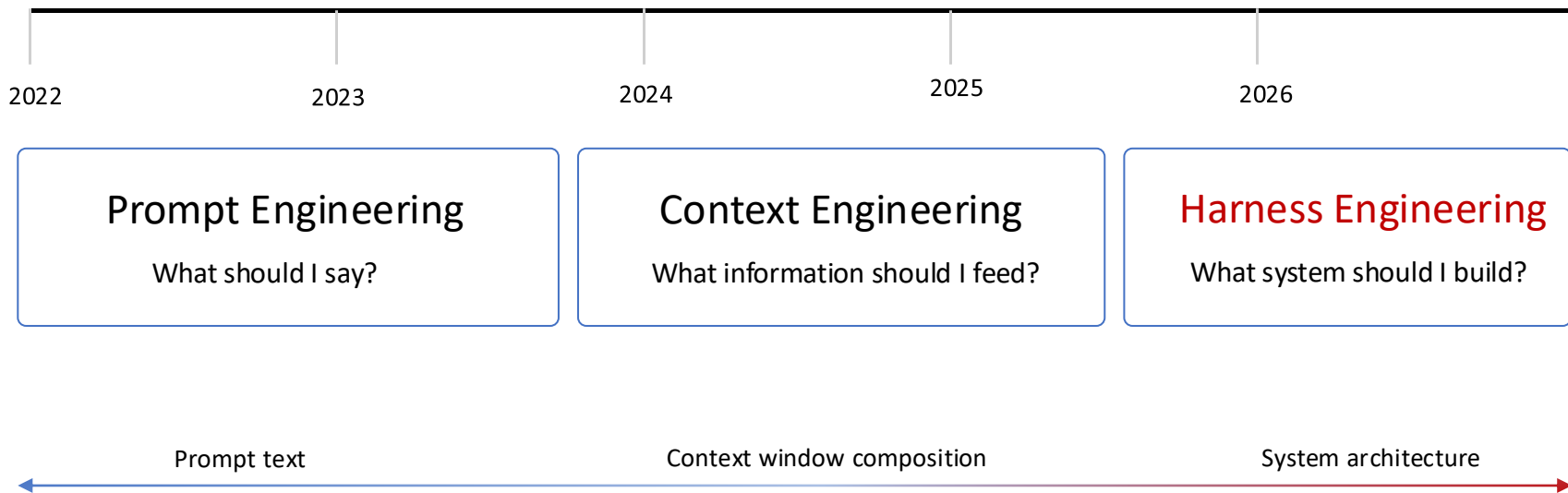
# Agenda

1. Evolution of AI Agentic Patterns
2. What is Harness Engineering?
3. Demo: Agent Harness with MAF and GitHub Copilot SDK



# Evolution of AI Agentic Patterns

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# What is Harness Engineering?

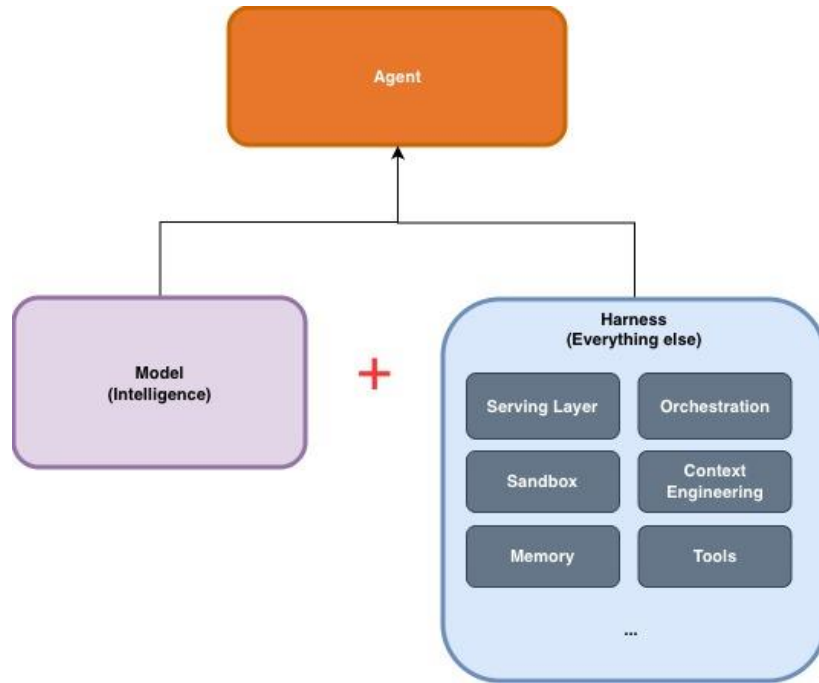
## What is Harness Engineering?

“Every time the agent makes a mistake, change the system so that mistake structurally cannot recur.”

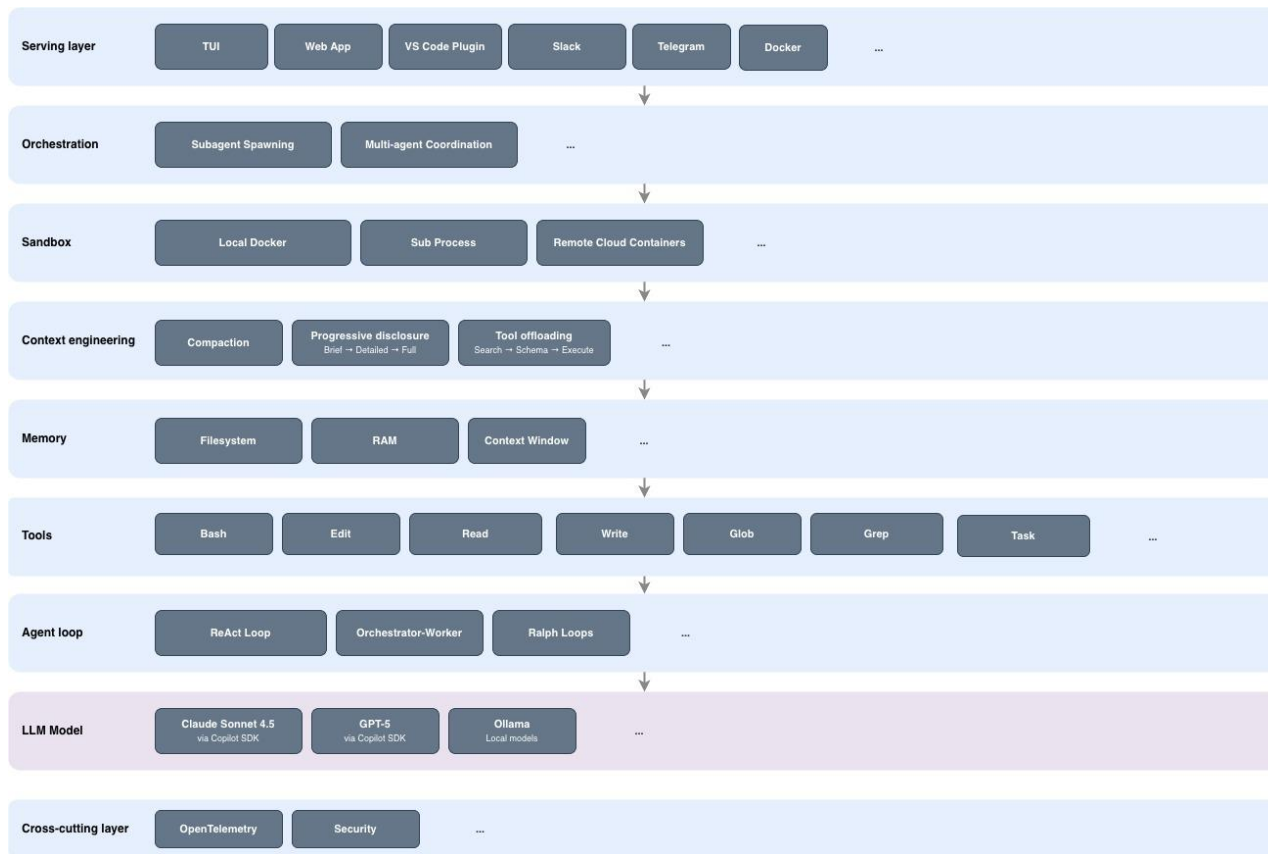
– Michell Hashimoto

- Not fixing the prompt. **Changing the system around the agent** — rules, tools, constraints, feedback loops  
=> The moment the term "harness engineering" was born.
- Fowler/Böckeler formula: Agent = Model + Harness

## Anatomy of an Agent Harness



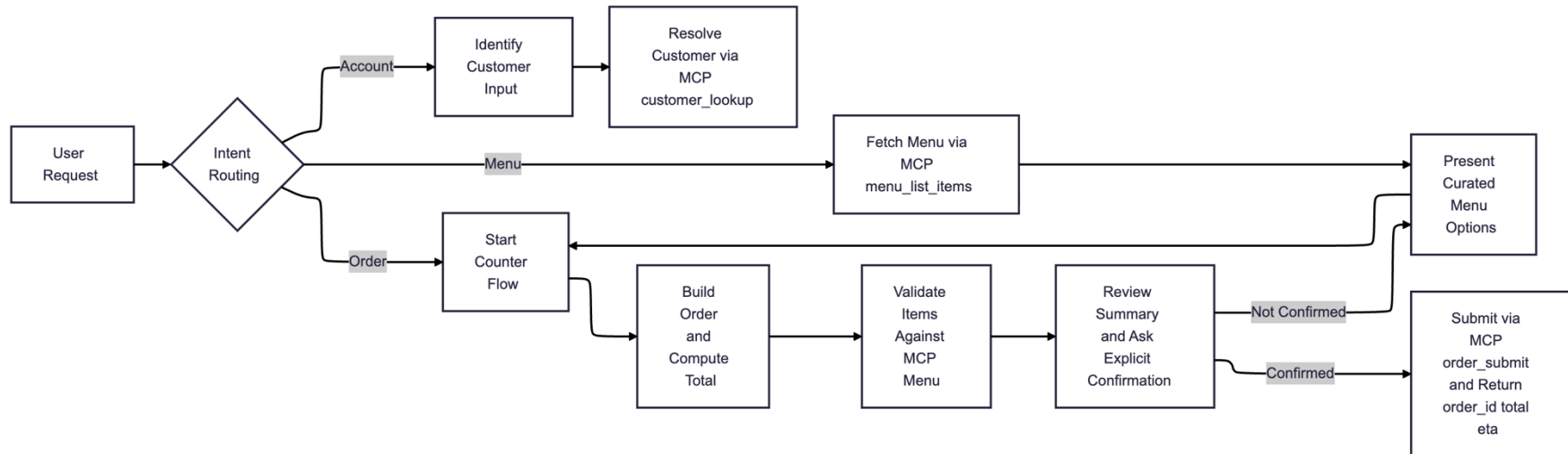
# Core components of an Agent Harness



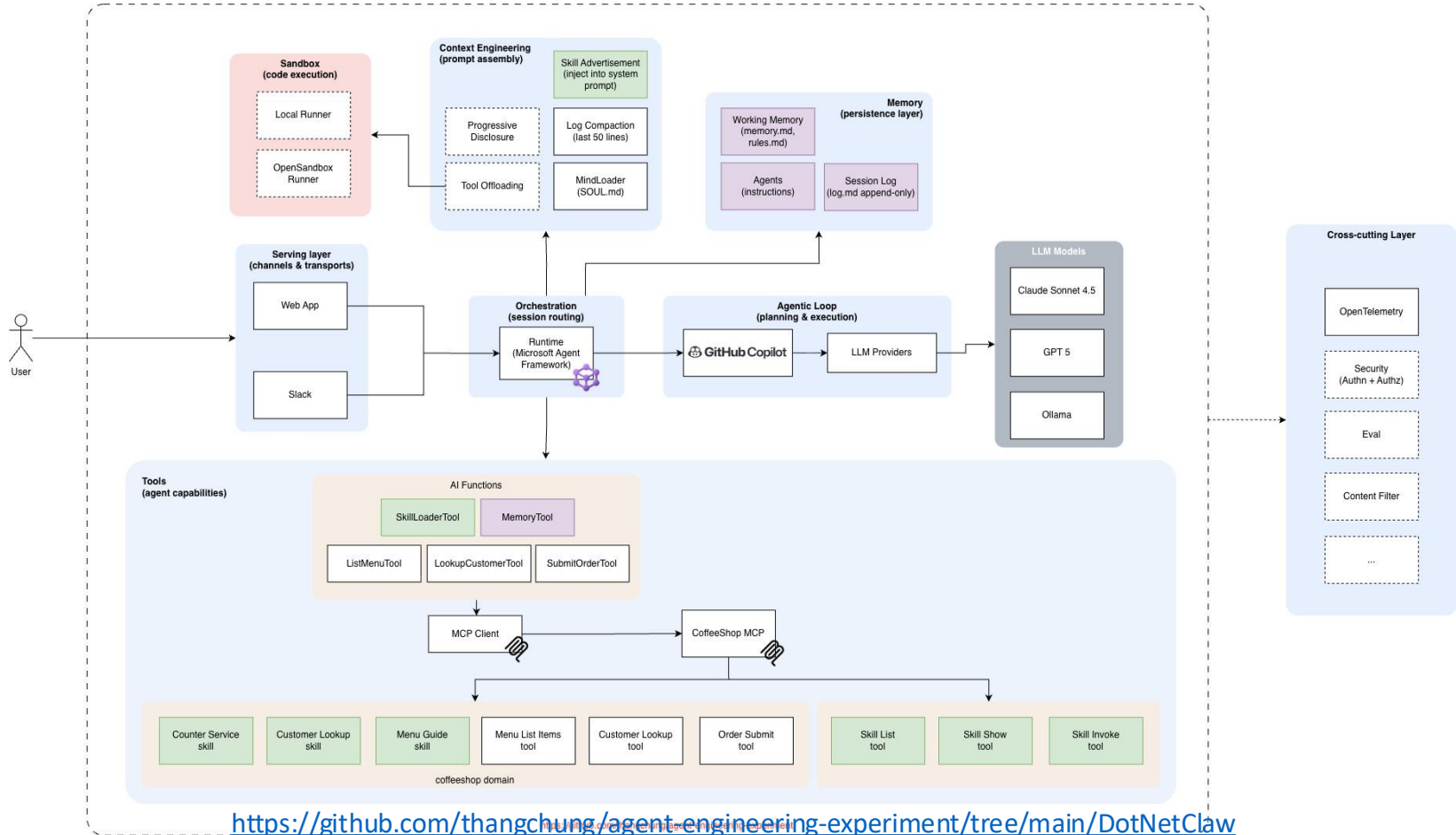


## Demo: Agent Harness with MAF and GitHub Copilot SDK

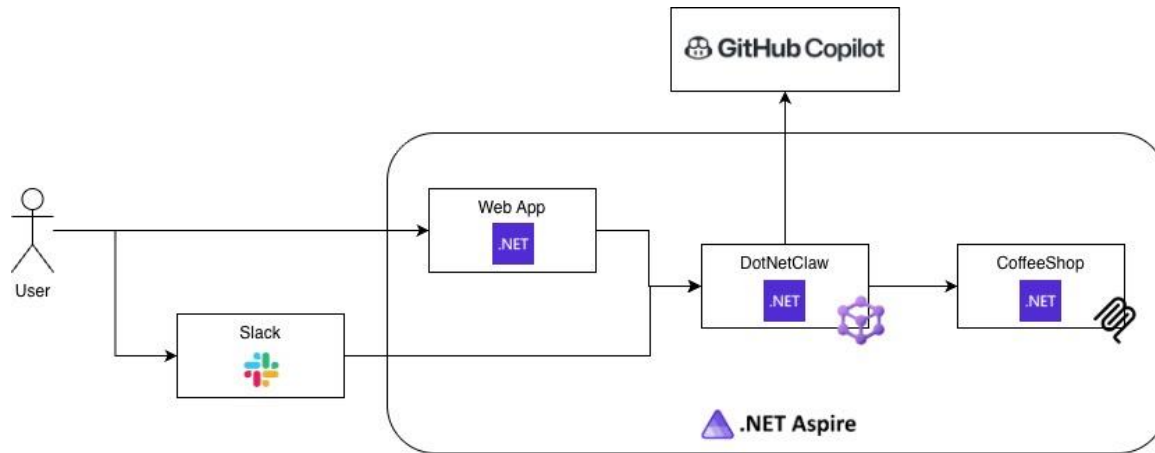
## DEMO: The coffeeshop agentic workflow (1/4)



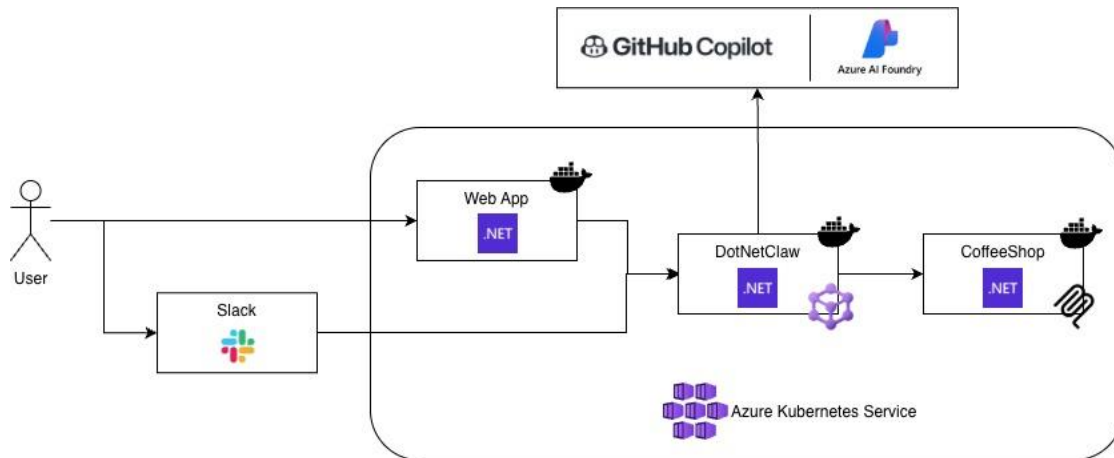
# DEMO: The coffeeshop agentic workflow (2/4)



## DEMO: Local dev and test (3/4)



## DEMO: Production with Azure Kubernetes Service (4/4)



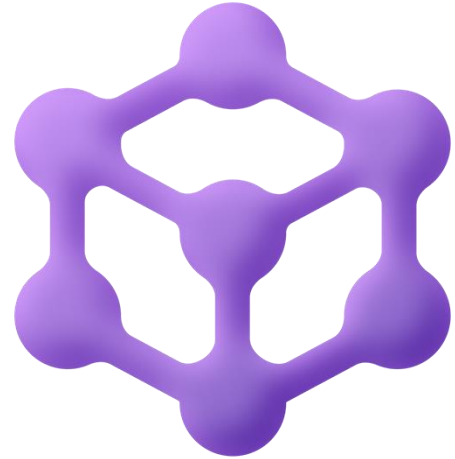


## Appendix: MAF and GitHub Copilot combination

# Microsoft Agent Framework (.NET/C#)

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- Microsoft Agent Framework (MAF) is a .NET/Python SDK for building AI agents with sessions, tools, memory, middleware, and multi-agent workflows.
- To standardize and harden agent development: structured orchestration, state management, model abstraction, and enterprise controls.
- Use it for tool-calling agents, multi-step workflows, or multi-agent systems.  
Do not use it for simple single LLM calls without orchestration.
- Link: <https://github.com/microsoft/agent-framework>

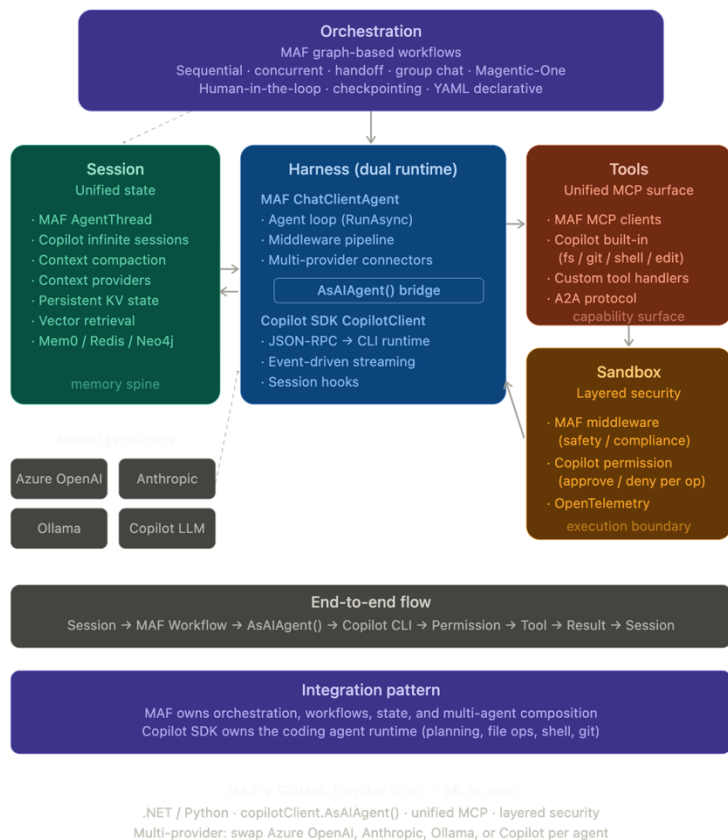


# GitHub Copilot SDK (.NET/C#)

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- GitHub Copilot SDK (.NET) is a library to embed the Copilot agent runtime into .NET applications with sessions, tool calling, and streaming.
- To reuse a production-ready agent loop instead of building your own model orchestration, tool dispatching, and conversation handling.
- Use it for multi-turn, tool-enabled agent workflows inside apps or services. Do not use it for simple one-shot model API calls.
- Link: <https://github.com/github/copilot-sdk>





## References

- <https://mitchellh.com/writing/my-ai-adoption-journey>
- <https://www.anthropic.com/engineering/managed-agents>
- <https://www.anthropic.com/engineering/effective-harnesses-for-long-running-agents>
- <https://www.humanlayer.dev/blog/skill-issue-harness-engineering-for-coding-agents>
- <https://blog.langchain.com/the-anatomy-of-an-agent-harness/>
- <https://martinfowler.com/articles/harness-engineering.html>
- <https://martinfowler.com/articles/exploring-gen-ai/harness-engineering.html>
- <https://openai.com/index/harness-engineering/>
- <https://bits-bytes-nn.github.io/insights/agent-ai/2026/04/05/evolution-of-ai-agent-patterns-en.html>
- <https://www.decodingai.com/p/agent-harness-engineering>
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Q&A